



Sydney Girls High School

2004
TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics

Extension 1

This is a trial paper ONLY.
It does not necessarily
reflect the format or the
contents of the 2004 HSC
Examination Paper in this
subject.

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Candidate Number

General Instructions

- Reading Time - 5 mins
- Working time - 2 hours
- Attempt ALL questions
- ALL questions are of equal value
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are supplied
- Board-approved calculators may be used.
- Diagrams are not to scale
- Each question attempted should be started on a new sheet. Write on one side of the paper only.

QUESTION 1 (12 marks)

Marks

- (a) Find the coordinates of the point that divides the interval joining $(-2, 5)$ and $(4, -3)$ externally in the ratio $3 : 2$

3

- (b) Solve $\frac{2x+1}{x-2} > 1$

3

- (c) Evaluate $\lim_{x \rightarrow 0} \left(\frac{\tan 4x}{\frac{x}{2}} \right)$

2

- (d) Use the substitution $u = 1 + x$ to evaluate $\int_0^1 \frac{x}{\sqrt{(1+x)^3}} dx$

4

Question 2. (12 marks)

Marks

- (a) Sketch the graph of $y=2\sin^{-1}\left(\frac{x}{2}\right)$

3

On your graph indicate the domain and range.

- (b) Differentiate and express in simplest form: $y=\sin^{-1}\left(\frac{x}{2}+1\right)$

3

- (c) Evaluate $\int_{-\frac{1}{3}}^{\frac{1}{3}} \frac{dx}{\sqrt{4-9x^2}}$

3

- (d) (i) Express $2 \sin \theta - \cos \theta$ in the form $A \sin(\theta - \alpha)$
where α is in radians, correct to 4 decimal places.

1

- (ii) Hence, or otherwise, solve for $0 \leq \theta \leq 2\pi$

2

$$2 \sin \theta - \cos \theta = \frac{\sqrt{5}}{2}$$

Give answer in radians correct to 2 decimal places.

Question 3. (12 marks)

Marks

- (a) A particle is moving in a straight line with its position (x) metres at time t seconds given by $x=2\cos\left(t+\frac{\pi}{4}\right)$.

(i) Show that the particle is moving in simple harmonic motion.

1

(ii) Write the period of its motion.

1

(iii) Find its maximum displacement.

1

(iv) Find its maximum velocity.

1

(v) Find the first time the particle is at the origin.

1

- (b) Prove by mathematical induction that $3^{2n} - 1$ is divisible by 8 for all integers $n \geq 1$

3

- (c) The area of a equilateral triangle of side length x cm is increasing at the rate of $2 \text{ cm}^2 \cdot \text{sec}^{-1}$

(i) Show that the area of the triangle is given by $A=\frac{\sqrt{3}}{4}x^2 \text{ cm}^2$

2

(ii) Find the exact rate of increase of the side (x) of the triangle when

$$x=2\sqrt{3} \text{ cm}$$

2

Question 4. (12 marks)

Marks

- (a) Taking $x = 0.6$ as a first approximation, use one application of Newton's method to find a second approximation to the root of $\tan x = x$ correct to 2 decimal places.

3

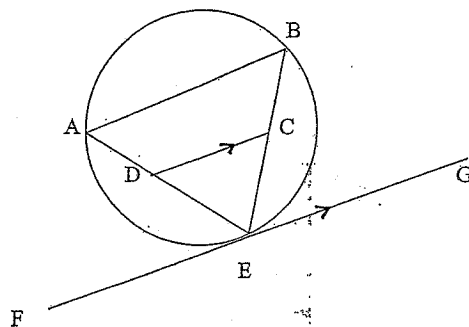
- (b) (i) Find the zeros of the polynomial function $P(x) = x^4 + 3x^3 + 2x^2$
(ii) Hence, without using calculus, sketch the polynomial function showing these zeros on the graph.

2

- (c) $DC \parallel FG$, FEG is a tangent at point E.

3

Copy this diagram and prove that A, B, C, D are concyclic points.



- (d) A particle moves with velocity $v = x - 5$ metres. sec^{-1}

If $x = 6$ metres initially

- (i) Show that the acceleration is the same as the velocity for all positions x

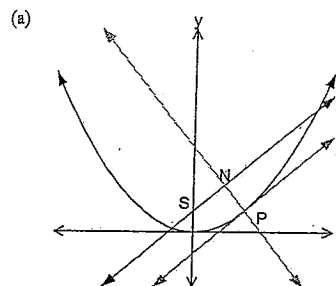
2

- (ii) Find x when $t = 4$ seconds

1

Question 5

Marks



The diagram shows the parabola $x^2=4ay$. P is the point $(2ap, ap^2)$

and $S(0,a)$ is the focus

- (i) Derive the equation of the normal to the parabola $x^2 = 4ay$ at the point P ($2ap$, ap^2)

1

- (ii) A line SN is drawn parallel to the tangent at P and intersects the normal PN at point N. Find the coordinates of point N.

2

- (iii) Show that the locus of N as P varies is a parabola and determine the vertex and focus of that parabola

3

- (b) Two of the roots of $x^3 - px^2 - qx - 20 = 0$ are 3 and 5

- (i) Find the other root

1

- (ii) Find p and q

2

- (c) (i) Find a general solution to $1 - 2 \cos 2x = 0$

2

- (ii) Sketch the curve $y = 1 - 2 \cos 2x$ for $0 \leq x \leq 2\pi$

1

Question 6 (12 marks)

Marks

- (a) Find $\int (\cos x + \sin x)^2 dx$ 3
- (b) (i) Find the domain and range of function $f: y = \frac{2}{x-1}$ 2
- (ii) Find the inverse function f^{-1} in terms of x . 1
- (iii) Sketch both functions on the same set of axes and state the co-ordinates of any common points. 3
- (c) From the top of a lighthouse 50 metres tall on a headland which is 750 metres above sea level, a tanker is seen on a bearing 320° at an angle of depression of 12° . A tugboat is also sighted at a bearing of 032° at an angle of depression of 20° . Calculate the distance between the vessels 3

Question 7

Marks

- (a) A skyrocket is fired from a height of 30 metres at an angle of 60° to the horizontal with a velocity of 20 ms^{-1} . Use $g = 10 \text{ ms}^{-2}$ to find in simplest exact form
- (i) The equations of motion for the horizontal and vertical components of displacement. 1
- (ii) The maximum height above ground that the rocket will reach. 2
- (iii) The total time the skyrocket is in flight. 2
- (iv) How far from the launching position will it land? 1
- (v) The speed at which the rocket hits the ground 2
- (b) The rate of change of temperature (T) for a substance cooling to room temperature (A) is given by
- $$\frac{dT}{dt} = -k(T-A) \quad \text{where } k \text{ and } A \text{ are constants}$$
- (i) Show that $T = A + Ce^{-kt}$ is a solution of this equation 1
- (ii) Initially the temperature is 130° . The temperature after one minute is 100° and after 2 minutes is 80° . Determine the room temperature

End of Exam

MATHS

Question 1

Ext 1 Trial 2004

$$\begin{aligned} \textcircled{3} \quad a \quad & \text{Externally } 3:2 \\ & = -3:2 \\ & = k_1:k_2 \\ x &= \frac{k_1 x_2 + k_2 x_1}{k_1 + k_2} \\ &= \frac{-3 \times 4 + 2 \times -2}{-3 + 2} \\ &= \frac{-16}{-1} \\ &= 16 \end{aligned}$$

$$\begin{aligned} y &= \frac{k_1 y_2 + k_2 y_1}{k_1 + k_2} \\ &= \frac{-3 \times -3 + 2 \times 5}{-3 + 2} \\ &= \frac{9 + 10}{-1} \\ &= -19 \end{aligned}$$

$\therefore$ Point = (16, -19)

$$\begin{aligned} \textcircled{2} \quad &= \lim_{x \rightarrow 0} \frac{4x}{\text{for } x/2} \\ &= 8 \lim_{x \rightarrow 0} \frac{x/2}{\text{for } x/2} \\ &= 8 \times 1 \\ &= 8 \end{aligned}$$

$$2) = 2\sqrt{2} + \sqrt{2} - 4$$

$$= 3\sqrt{2} - 4$$

$$\begin{aligned} \textcircled{3} \quad b \quad & \frac{2x+1}{x-2} > 1 \\ \text{1st C.V. } x &= 2 \\ \text{Let } \frac{2x+1}{x-2} &= 1 \\ 2x+1 &= x-2 \\ x &= -3 \quad \text{2nd C.V.} \end{aligned}$$

$$\frac{-3}{-3} \quad \frac{2}{2}$$

Test $x = -4$ in ineq.

$$\frac{-8+1}{-4-2} = \frac{-7}{-6} = 1\frac{1}{6} > 1$$

Test $x = 0$ in ineq.

$$\frac{1}{-2} \not> 1$$

Test $x = 3$ in ineq.

$$\frac{6+1}{3-2} = \frac{7}{1} > 1$$

Ans: $x < -3$, $x > 2$

$$\begin{aligned} \textcircled{4} \quad & \text{Let } u = 1+x \\ & du = dx \\ x=0, u &= 1 \\ x=1, u &= 2 \\ \int_0^1 \frac{x \, dx}{\sqrt{(1+x)^3}} &= \int_1^2 \frac{u-1}{u^{3/2}} \, du \\ &= \int_1^2 u^{-1/2} - u^{-3/2} \, du \\ &= \left[2u^{1/2} + 2u^{-1/2} \right]_1^2 \\ &= 2 \left[\sqrt{u} + \frac{1}{\sqrt{u}} \right]_1^2 = 2 \left[\sqrt{2} + \frac{1}{\sqrt{2}} - 2 \right] \end{aligned}$$

Question 2

$$\textcircled{3} \quad a \quad y = 2 \sin^{-1} \left(\frac{x}{2} \right)$$

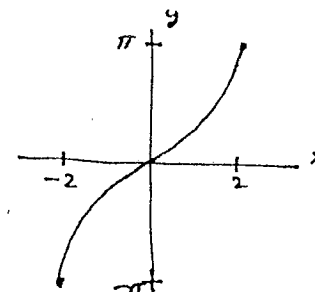
$$-1 \leq \frac{x}{2} \leq 1$$

$$\therefore -2 \leq x \leq 2 \quad \text{Dom}$$

$$-\frac{\pi}{2} \leq \sin^{-1} \left(\frac{x}{2} \right) \leq \frac{\pi}{2}$$

$$-\pi \leq 2 \sin^{-1} \frac{x}{2} \leq \pi$$

$$\therefore -\pi \leq y \leq \pi \quad \text{Range}$$



$$\begin{aligned} \textcircled{3} \quad b \quad & y = \sin^{-1} \left(\frac{x}{2} + 1 \right) \\ & \text{Let } u = \frac{x}{2} + 1 \\ & \frac{du}{dx} = \frac{1}{2} \\ \frac{dy}{dx} &= \frac{1}{\sqrt{1-u^2}} \cdot \frac{1}{2} \\ &= \frac{1}{2 \sqrt{1 - (\frac{x}{2} + 1)^2}} \\ &= \frac{1}{2 \sqrt{-x^2 - 4x}} \\ &= \frac{1}{\sqrt{-x^2 - 4x}} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad &= \int_{-\frac{1}{3}}^{\frac{1}{3}} \frac{dx}{\sqrt{4-9x^2}} \\ &= \int_{-\frac{1}{3}}^{\frac{1}{3}} \frac{dx}{3 \sqrt{\frac{4}{9} - x^2}} \\ &= \frac{1}{3} \int_{-\frac{1}{3}}^{\frac{1}{3}} \frac{dx}{\sqrt{(\frac{2}{3})^2 - x^2}} \\ &= \frac{1}{3} \left[\sin^{-1} \left(\frac{3x}{2} \right) \right]_{-\frac{1}{3}}^{\frac{1}{3}} \\ &= \frac{1}{3} \left[\left(\sin^{-1} \frac{1}{2} \right) - \left(\sin^{-1} \left(-\frac{1}{2} \right) \right) \right] \\ &= \frac{1}{3} \left[\frac{\pi}{6} + \frac{\pi}{6} \right] \\ &= \frac{\pi}{9} \end{aligned}$$

Question 2 (cont)

① i $2 \sin \theta - \cos \theta$

$$A \sin(\theta - \alpha) = A \sin \theta \cos \alpha - A \cos \theta \sin \alpha$$

$$= (A \cos \alpha) \sin \theta - (A \sin \alpha) \cos \theta$$

$$A \cos \alpha = 2 \quad A \sin \alpha = 1$$

$$A^2 \cos^2 \alpha + A^2 \sin^2 \alpha = 5$$

$$A^2 (\cos^2 \alpha + \sin^2 \alpha) = 5$$

$$A = \sqrt{5}$$

$$\frac{A \sin \alpha}{A \cos \alpha} = \frac{1}{2}$$

$$\tan \alpha = \frac{1}{2}$$

$$\therefore \alpha = \tan^{-1} \frac{1}{2} = 0.4636 \quad (4 \text{ d.p.})$$

$$\text{Hence } 2 \sin \theta - \cos \theta = \sqrt{5} \sin(\theta - 0.4636)$$

② ii Solve for $0 \leq \theta \leq 2\pi$

$$2 \sin \theta - \cos \theta = \frac{\sqrt{5}}{2}$$

$$\sqrt{5} \sin(\theta - 0.4636) = \frac{\sqrt{5}}{2}$$

$$\sin(\theta - 0.4636) = \frac{1}{2}$$

$$\theta - 0.4636 = 0.5236 \quad \text{or} \quad 2.6180$$

$$\text{Ans } \theta = 3.08 \quad \text{or} \quad 0.99$$

Question 3

① i $x = 2 \cos(t + \frac{\pi}{4})$

$$\dot{x} = -2 \sin(t + \frac{\pi}{4})$$

$$\ddot{x} = -2 \cos(t + \frac{\pi}{4})$$

$$\therefore \ddot{x} = -1x \quad \text{which is in form } \ddot{x} = -\omega^2 x$$

$$\therefore \text{S.H.M.}$$

① ii

$$A = 1$$

$$\therefore \omega = 1$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{1} = 2\pi \text{ secs.}$$

① iii

$$x = a \cos(\omega t + \alpha)$$

$$\therefore a = 2 \text{ m} = \text{Maximum displacement.}$$

① iv

Maximum velocity occurs at $x = 0$

$$v^2 = \omega^2 (a^2 - x^2)$$

$$= 1 (4 - 0)$$

$$\therefore v = 2 \text{ m s}^{-1} = \text{Max. velocity.}$$

① v

Let $x = 0$

$$0 = 2 \cos(t + \frac{\pi}{4})$$

$$\cos(t + \frac{\pi}{4}) = 0$$

$$\therefore t + \frac{\pi}{4} = \frac{\pi}{2}$$

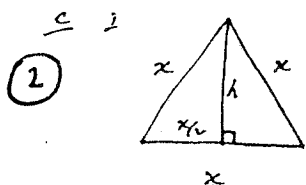
$$\therefore t = \frac{\pi}{4} = \text{1st time the particle is at origin.}$$

Question 3

- Q1) Prove true for $n=1$
 Let $n=1$, $3^{2n}-1 = 8$ which is divisible by 8
 $\therefore$ True for $n=1$
 Assume true for $n=k$
 $\therefore 3^{2k}-1 = 8m$ which is divisible by 8
 Prove true for $n=k+1$

$$\begin{aligned} 3^{2(k+1)} - 1 &= 3^{2k+2} - 1 = 3^{2k} \cdot 3^2 - 1 \\ &= (3^{2k} - 1)9 + 8 \\ &= (8m)9 + 8 \\ &= 8(9m+1) \text{ which is divisible by 8} \end{aligned}$$

If true for $n=k$, then true for $n=k+1$
 Since true for $n=1$, then true for $n=2, n=3$ etc
 Hence by math. induction, true for all $n \geq 1$



Data $\frac{dA}{dt} = 2 \text{ cm/s}$

$$\begin{aligned} h^2 &= x^2 - \frac{x^2}{4} = \frac{3x^2}{4} \\ h &= \frac{\sqrt{3}x}{2} \end{aligned}$$

$$\begin{aligned} \therefore A &= \frac{1}{2} \times x \times \frac{\sqrt{3}x}{2} \\ &= \frac{\sqrt{3}x^2}{4} \end{aligned}$$

ii

$$\begin{aligned} \frac{dA}{dx} &= \frac{2\sqrt{3}x}{4} \\ &= \frac{\sqrt{3}x}{2} \end{aligned}$$

$$\begin{aligned} \frac{dA}{dt} &= \frac{dA}{dx} \cdot \frac{dx}{dt} \\ 2 &= \frac{\sqrt{3}x}{2} \cdot \frac{dx}{dt} \end{aligned}$$

$$\begin{aligned} \therefore \frac{dx}{dt} &= \frac{4}{\sqrt{3}} \cdot \frac{1}{x} \quad \text{put } x = 2\sqrt{3} \\ &= \frac{4}{\sqrt{3}} \cdot \frac{1}{2\sqrt{3}} \\ &= \frac{2}{3} \text{ cm/s} = \text{Rate of increase of side.} \end{aligned}$$

Question 4

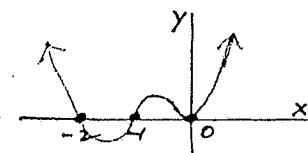
- Q3) a $\tan x = x$
 $\tan x - x = 0$
 $\therefore f(x) = \tan x - x$
 $f'(x) = \sec^2 x - 1$

Let $x = 0.6$
 $a_1 = x - \frac{f(x)}{f'(x)}$
 $= 0.6 - \frac{\tan(0.6) - 0.6}{\sec^2(0.6) - 1}$
 $= 0.42 = 2^{\text{nd}} \text{ approx to } x$

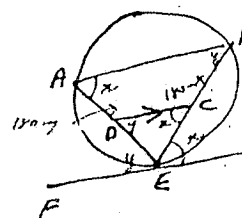
- b i $P(x) = x^4 + 3x^3 + 2x^2$
 $= x^2(x^2 + 3x + 2)$
 $= x^2(x+2)(x+1)$
 $\therefore$ Zeros are $x = 0, -2, -1$
 (double root)

ii Test $x = 1$, $P(1) = 1 + 3 + 2 = 6$

Q1



Q3



Aim: Prove A, B, C, D concyclic

Proof: $\angle DEB = \angle DAE$ (L in alt seg.)
 $\angle BEG = \angle DCE$ (alt L's, $DE \parallel FG$)
 $= x$

$\angle DCB = 180 - \angle DCE = 180 - x$ (L's st line)

$\therefore \angle DAB + \angle DCB = 180^\circ$

$\angle AEF = \angle ABE = y$ (L in alt seg.)

$\angle BEF = \angle CDE = x$ (alt L's, $DE \parallel FG$)

$\angle ADC = 180 - \angle CDE = 180 - y$ (L's st line)

$\therefore \angle ABC + \angle ADC = 180^\circ$

Hence A, B, C, D are concyclic points (opp L's are supp.)

Question 4

(2) i $V = x - 5$
 $V^2 = (x - 5)^2$
 $= x^2 - 10x + 25$
 $\frac{1}{2} V^2 = \frac{1}{2} x^2 - 5x + \frac{25}{2}$
 $acc = \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} V^2 \right)$
 $= x - 5$

Since $\ddot{x} = V \therefore$ acc is the same as vel for all x

(2) ii $V = \frac{dx}{dt} = x - 5$
 $\frac{dx}{x - 5} = dt$
 $\int dt = \int \frac{dx}{x - 5}$

$t = \log_e (x - 5) + c$

$t = 0, x = 6 \therefore 0 = \log_e (6 - 5) + c$
 $0 = \log_e 1 + c$
 $\therefore c = 0$

$t = \log_e (x - 5)$

Put $t = 4 \quad 4 = \log_e (x - 5)$

$\therefore x - 5 = e^4$
 $x = e^4 + 5$ metres
 $= 59.6 \text{ m}$

Part a. $t = 0, x = 6$

Q5 b $x^3 - px^2 - qx - 20 = 0$

i Let roots be $2, 3, 5$

ii $p = 13 + 5 = 9$
 $q = 3\left(\frac{2}{3}\right) + 5\left(\frac{3}{5}\right) + 15$
 $= 25$

(1)

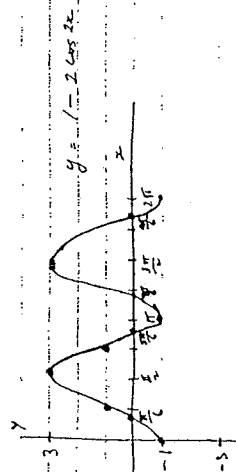
iii $1 - 2 \cos 2\alpha = 0$

$2 \cos 2\alpha = 1$
 $\cos 2\alpha = \frac{1}{2}$

$2\alpha = \cos^{-1} \frac{1}{2} = \frac{\pi}{3}$

General Soln: $2\alpha = 2n\pi \pm \frac{\pi}{3}$
 $\therefore \alpha = n\pi \pm \frac{\pi}{6}$ ($n = \text{any integer}$)

ii



(1)

Question 5

i $x = 2p$
 $y = 4p$
 $\frac{dy}{dx} = 2$
 $\therefore$ gradient = $-\frac{1}{2}p$

ii $xy - y^2 = -x + 2y$
 $\therefore x + y^2 = 2y$

iii $6x^2 + 5y = p$
 $y = \frac{p}{5} - \frac{6}{5}x^2$
 $\therefore y = px + a$

Solve Sim $y = px + a$
 $x + \left(\frac{6}{5}x^2 + \frac{p}{5}\right) = 2\left(\frac{6}{5}x^2 + \frac{p}{5}\right)$
 $x(1 + \frac{6}{5}x) = \frac{p}{5}(1 + \frac{6}{5}x)$
 $x = \frac{p}{5}$

$\therefore$ Point N = $\left(\frac{p}{5}, \frac{p}{5}\right)$

iii $x = ap$
 $y = a(p^2 + 1)$
 $\therefore \left(\frac{x^2}{a^2} - 1\right)$

$y = \frac{x^2}{a} + a$

$\therefore x = a(y - a)$ is focus of N which is a parabola.

Vertex = $(0, a)$ Focus = $\left(0, \frac{5a}{4}\right)$

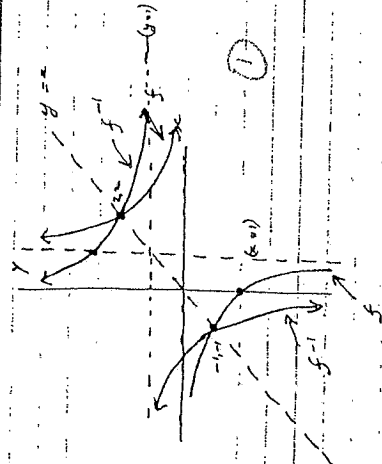
Question 6

6. i $f: y = \frac{2}{x-1}$ Dom = all real x but $x \neq 1$
Range = all real y but $y \neq 0$

ii $f^{-1}: x = \frac{2}{y-1}$

$y-1 = \frac{2}{x}$
 $y = \frac{2}{x} + 1$

iii Dom: all real x but $x \neq 0$
Range: all real y but $y \neq 1$

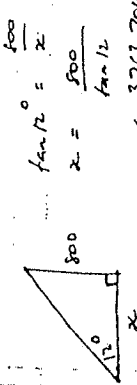
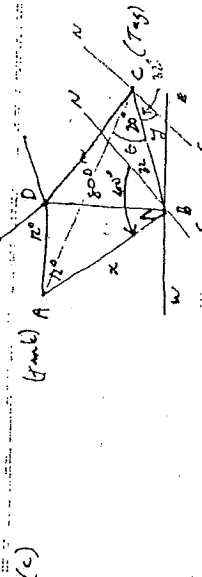


For common points Solve $\begin{cases} y = \frac{2}{x-1} \\ y = x \end{cases}$

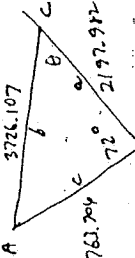
$\frac{2}{x-1} = x$
 $2 = x(x-1)$
 $0 = x^2 - x - 2$
 $0 = (x-2)(x+1)$
 $\therefore (x=2, y=2)$ and $(x=-1, y=-1)$

$\int (\cos x + \sin x) dx$
 $= \int \cos x dx + \int \sin x dx$
 $= \sin x - \cos x + c$

3



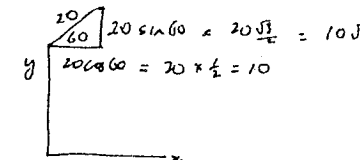
$\tan 72^\circ = \frac{800}{x}$
 $x = \frac{800}{\tan 72^\circ}$
 $x = 3763.704$



Cosine Rule
 $6^2 = a^2 + b^2 - 2ab \cos B$
 $6 = 3763.107$
Distance between vehicles = 3726 metres

3

87



$g = 10$

Vert.
 i $t=0, y = 10\sqrt{3}, y = 30$

$y'' = -g$
 $y' = -gt + c$
 $10\sqrt{3} = c$
 $y = -\frac{gt^2}{2} + 10\sqrt{3}t + c$
 $30 = c$
 $y = -5t^2 + 10\sqrt{3}t + 30$

Horiz.
 $t=0, x=0, x=10$

$x'' = 0$
 $x' = 0 + c$
 $10 = c$
 $x = 10t + c$
 $x = 10t$

1

ii At max height $y' = 0$
 $0 = -10t + 10\sqrt{3}$
 $t = \sqrt{3}$
 $\therefore$ Max Height = y
 $= -5 \times 3 + 10\sqrt{3} \times \sqrt{3} + 30$
 $= -15 + 30 + 30$
 $= 45$ metres

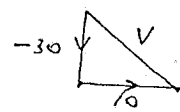
iii Let $y = 0$
 $5t^2 - 10\sqrt{3}t - 30 = 0$
 $t^2 - 2\sqrt{3}t - 6 = 0$
 $t = \frac{2\sqrt{3} \pm \sqrt{12 + 24}}{2}$
 $= \frac{2\sqrt{3} + 6}{2}$
 $= \sqrt{3} + 3$ s
 $=$ total flight time.

2

iv $x = 10 \times t$
 $= 10(\sqrt{3} + 3)$ metres
 $=$ range.

1

$\therefore$ Let $t = \sqrt{3} + 3$
 $y' = -10(\sqrt{3} + 3) + 10\sqrt{3} = -30$
 $x = 10$



$V^2 = (-30)^2 + 10^2$
 $= 900 + 100$
 $= 1000$
 $V = \sqrt{1000}$
 $= 10\sqrt{10}$ m/s
 $=$ speed it hits the ground

2

Q 7b : $T = A + Ce^{-kt}$

$$\frac{dT}{dt} = -kCe^{-kt}$$

$$-k(T-A) = -k(Ce^{-kt}) \quad \therefore \frac{dT}{dt} = -k(T-A)$$

Hence $T = A + Ce^{-kt}$ is a solution. (1)

ii) $t=0, T=130$

$$130 = A + C$$

$t=1, T=100$

$$100 = A + Ce^{-k}$$

$t=2, T=80$

$$80 = A + Ce^{-2k}$$

$$A = 130 - C$$

$$100 = 130 - C + Ce^{-k}$$

$$\therefore C(1 - e^{-k}) = 30 \quad (1)$$

$$80 = 130 - C + Ce^{-2k}$$

$$\therefore C(1 - e^{-2k}) = 50 \quad (2)$$

$$(1) \div (2)$$

$$\frac{1 - e^{-k}}{1 - e^{-2k}} = \frac{3}{5} \quad (3)$$

$$\frac{1-m}{1-m^2} = \frac{3}{5}$$

$$\text{Put } e^{-k} = m \quad \therefore e^{-2k} = m^2$$

$$5 - 5m = 3 - 3m^2$$

$$3m^2 - 5m + 2 = 0$$

$$(3m-2)(m-1) = 0$$

$$\therefore m = \frac{2}{3} \quad \text{or} \quad m = 1$$

$$\text{Now } e^{-k} = \frac{2}{3} \quad \text{or} \quad e^{-k} = 1$$

$$\text{Taking } e^{-k} = 1$$

$$\text{means (3) becomes } \frac{1-1}{1-1} = \frac{0}{0} \neq \frac{3}{5}$$

$$\therefore e^{-k} = \frac{2}{3} \quad \text{only}$$

Sub into (1)

$$C = \frac{30}{(1 - \frac{2}{3})} = 90$$

$$\therefore A = 130 - 90 = 40$$

Thus Room Temp = 40° (3)